

Interacting Multipoles of Rank 2 and 3:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{aligned} &-945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta(\delta, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta(\gamma, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta(\beta, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &+105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \gamma) \cdot \delta(\delta, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \delta) \cdot \delta(\gamma, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \epsilon) \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \gamma) \cdot \delta(\delta, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \delta) \cdot \delta(\gamma, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \epsilon) \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \beta) \cdot \delta(\delta, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \epsilon) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \epsilon) \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ &-15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \end{aligned} \right] \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{array}{l} -945 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot \delta(\delta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\delta\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\epsilon \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\gamma\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot R_\epsilon \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\delta} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\epsilon} \\ +105 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\delta\epsilon} \\ +105 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\delta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \delta(\delta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\alpha \cdot \delta(\beta, \beta) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\beta\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \delta(\delta, \delta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\beta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\gamma\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \delta(\delta, \delta) \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\delta\delta} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\gamma} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\gamma\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\delta} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\gamma, \gamma) \cdot \Theta_{\alpha\alpha} \cdot \Omega_{\gamma\gamma\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\epsilon} \\ -15 \cdot R_z^4 \cdot R_\epsilon \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Theta_{\alpha\beta} \cdot \Omega_{\alpha\beta\epsilon} \end{array} \right] \quad (2a)$$

Simplify R:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{aligned} &-945 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Omega_{zzz} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Omega_{z\delta\delta} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Omega_{z\gamma\gamma} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{zz} \cdot \Omega_{z\gamma\gamma} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ &+105 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Omega_{zzz} \\ &-15 \cdot R_z^4 \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{\beta\delta\delta} \\ &-30 \cdot R_z^4 \cdot R_z \cdot \Theta_{z\beta} \cdot \Omega_{\beta\gamma\gamma} \\ &-15 \cdot R_z^4 \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\delta\delta} \\ &-30 \cdot R_z^4 \cdot R_z \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ &-15 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Omega_{z\delta\delta} \\ &-30 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &-15 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Omega_{z\gamma\gamma} \\ &-30 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &-15 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\alpha} \cdot \Omega_{z\gamma\gamma} \\ &-30 \cdot R_z^4 \cdot R_z \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \end{aligned} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{aligned} &+105 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ &+105 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ &+105 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ &+105 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ &+105 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ &+105 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ &-15 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{\beta\delta\delta} \\ &-30 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{\beta\gamma\gamma} \\ &-15 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\delta\delta} \\ &-30 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ &-30 \cdot R_z^5 \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &-30 \cdot R_z^5 \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \\ &-30 \cdot R_z^5 \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \end{aligned} \right] \quad (3b)$$

$$\xrightarrow{\text{added up}} R_z^{-11} \cdot -\frac{1}{45} \cdot \left[\begin{aligned} &+315 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{zz\beta} \\ &+315 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{zz\alpha} \\ &-15 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{\beta\delta\delta} \\ &-30 \cdot R_z^5 \cdot \Theta_{z\beta} \cdot \Omega_{\beta\gamma\gamma} \\ &-15 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\delta\delta} \\ &-30 \cdot R_z^5 \cdot \Theta_{z\alpha} \cdot \Omega_{\alpha\gamma\gamma} \\ &-90 \cdot R_z^5 \cdot \Theta_{\alpha\beta} \cdot \Omega_{z\alpha\beta} \end{aligned} \right] \quad (3c)$$

Simplify Quadrupoles:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = R_z^{-11} \cdot -\frac{1}{45} \cdot \begin{bmatrix} +315 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{zzz} \\ +315 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{zzz} \\ -15 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{z\delta\delta} \\ -30 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{z\gamma\gamma} \\ -15 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{z\delta\delta} \\ -30 \cdot R_z^5 \cdot \Theta_{zz} \cdot \Omega_{z\gamma\gamma} \\ -90 \cdot R_z^5 \cdot \Theta_{\alpha\alpha} \cdot \Omega_{z\alpha\alpha} \end{bmatrix} \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-11} \cdot -\frac{1}{45} \cdot [+0] \quad (4b)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon}\Theta_{\alpha\beta}\Omega_{\gamma\delta\epsilon} = +0 \quad (5)$$